

Lesson 20 — Design Challenge — Systems Demo Day

Matches LEGO Teacher Portal lesson: "Game Designer".

Objective (I can...): I can design a stateful system that stores, updates, and reasons over data to drive behavior. **CS concept:** Applying variables with events, loops, and conditionals to a stateful build · **Builds on:** Lessons 16-19. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): An adaptive game AI tracks your performance and adjusts difficulty — data drives its choices. *Impacts prompt:* "Adaptive systems can help some players and frustrate others — what's a fair way to design that?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, trigger objects, notebooks. **Model already built** (Variables model) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Variables review — create, update, use-in-decision, multiple, state, derived value. (*Theme: Systems demo day.*) - **Concept + demo (10):** Challenge: build a stateful system (scorekeeper, timer game, resource meter) that uses at least one variable plus an event, a loop, and a conditional, and clearly changes state. Coach models the plan, then hands off. - **Build/Code (30):** Pairs plan and code the stateful system on the built model, documenting each variable's role in the notebook. *Swap at 15 min.* - **Challenge (7):** Add a second variable, a reset, or a win/lose condition that compares two variables. - **Share + pack-up (5): Systems demo day** — pairs run their system live while another pair tries to make it change state; creators explain the variables behind the behavior. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the variable logic; swap at 15 min. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a counter-and-threshold target with a partial plan. *Extension / fast-finisher* — require two variables, a derived value, and a win condition. *Reteach* — reuse a Lesson 18-19 program as a template.

Exit check: Pair delivers a working stateful system that stores, updates, and uses variables to drive behavior, with each variable documented. **Progress layer:** Send home the **Unit 4 parent note** ("your student can build a system that tracks and reasons over data") with the **build photo**. **At-home (optional):** Design a simple game on paper; list every variable it needs and when each changes. ~10 min.